Negative results and claim revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Negative results and claim revision before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Robustness across conditions

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Running the primary experiment

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Negative results and claim revision trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Negative result**s and **claim revision**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Negative results and claim revision explanation against the supplied facts before discussing it.

Use the worked case for **Negative result**s and **claim revision**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Hypothesis predicts mean latency reduction at least2 seconds. Observed paired mean reduction .8. Report.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Negative results and claim revision facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Negative results and claim revision facts and keep your reasoning.

Hypothesis expects B support strictly above A. Both pass8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Negative results and claim revision facts and keep your reasoning.

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact **equivalence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Negative results and claim revision facts and keep your reasoning.

Hypothesis predicts mean latency reduction at least2 seconds. Observed paired mean reduction .8. Report.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Negative results and claim revision facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Negative results and claim revision to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **negative result**s and **claim revision evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **negative result**s and **claim revision**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Negative results and claim revision

> A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation. Explain the labelled step before checking the changed condition.

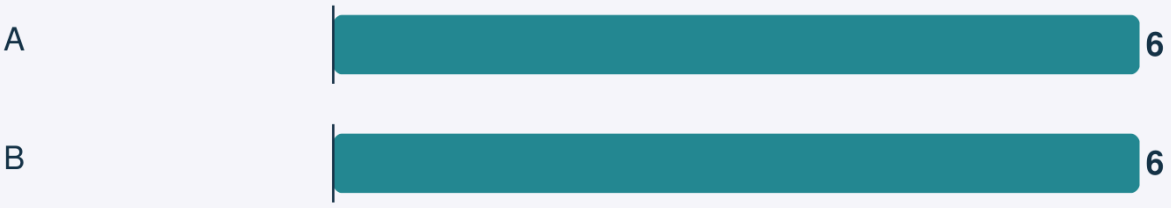
A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

A and B each score 6/8. Before running, the team predicted B would score higher.

AI ACADEMY / AI-7/28

No observed count advantage

correct cases out of 8 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/28

Negative results and claim revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Recall the prediction

The claim was a higher correct count for B.

↓

2. State the finding

Both versions score 75%; no observed count advantage appears here.

↓

3. Revise cautiously

Report the unsupported prediction and a justified next test.

CHANGE ONE THING

B wins on one new task after the team chooses it because A fails.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Does a 6/8 tie support the **prediction** that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the unsupported **prediction** and a justified next test.

1. Revise cautiously

2. Recall the **prediction**

3. State the finding

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Revise cautiously Report the unsupported **prediction** and a justified next test. A **negative result** can be informative when methods and limits are clear.

A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: B wins on one new task after the team chooses it because A fails. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Treat that selected demonstration as exploratory, not a replacement for the original result.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Treat that selected demonstration as exploratory, not a replacement for the original result. Case selection now depends on outcomes.

Repair a missing building block

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

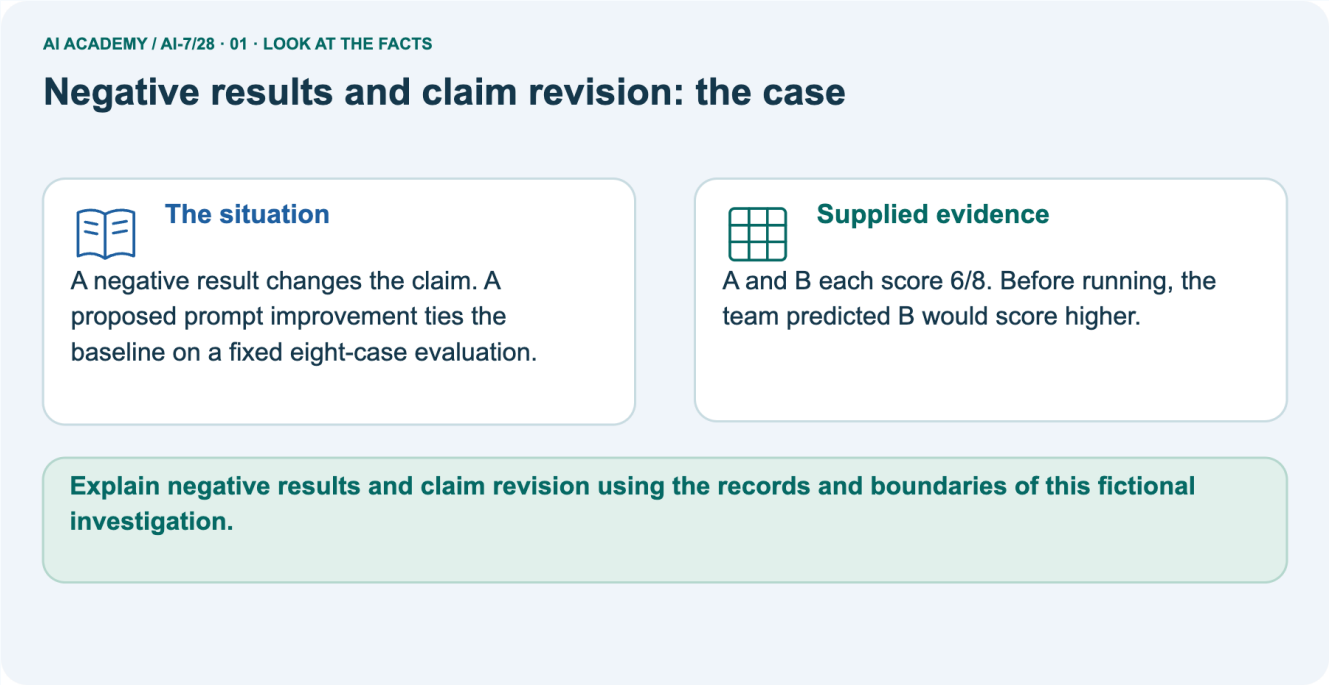
> **What I repaired, the evidence I kept and the next test**

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Negative results and claim revision picture from its labels, then record a relevant limit.

01. Negative results and claim revision: the case



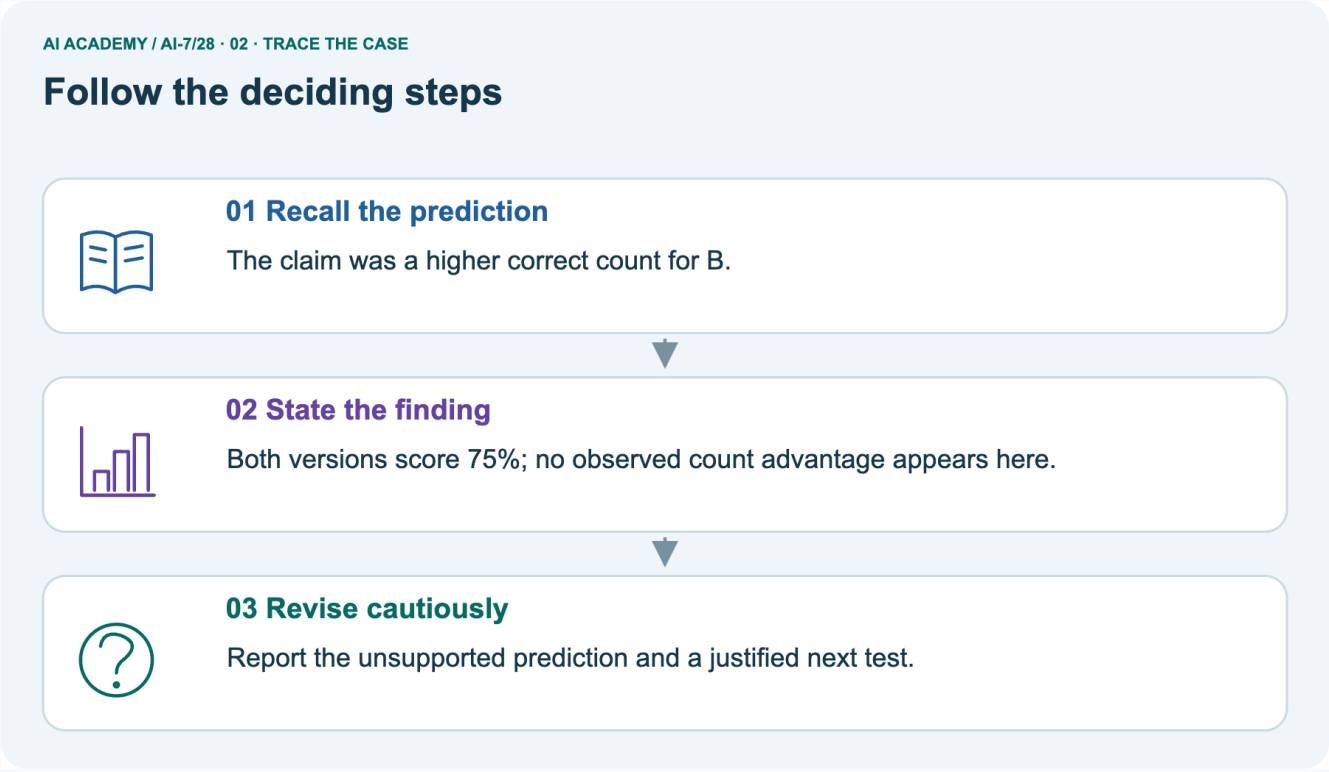
Follow the labelled picture with the matching explanation. Open the vector diagram

A **negative result** changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation. Supplied facts: A and B each score 6/8. Before running, the team predicted B would score higher.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



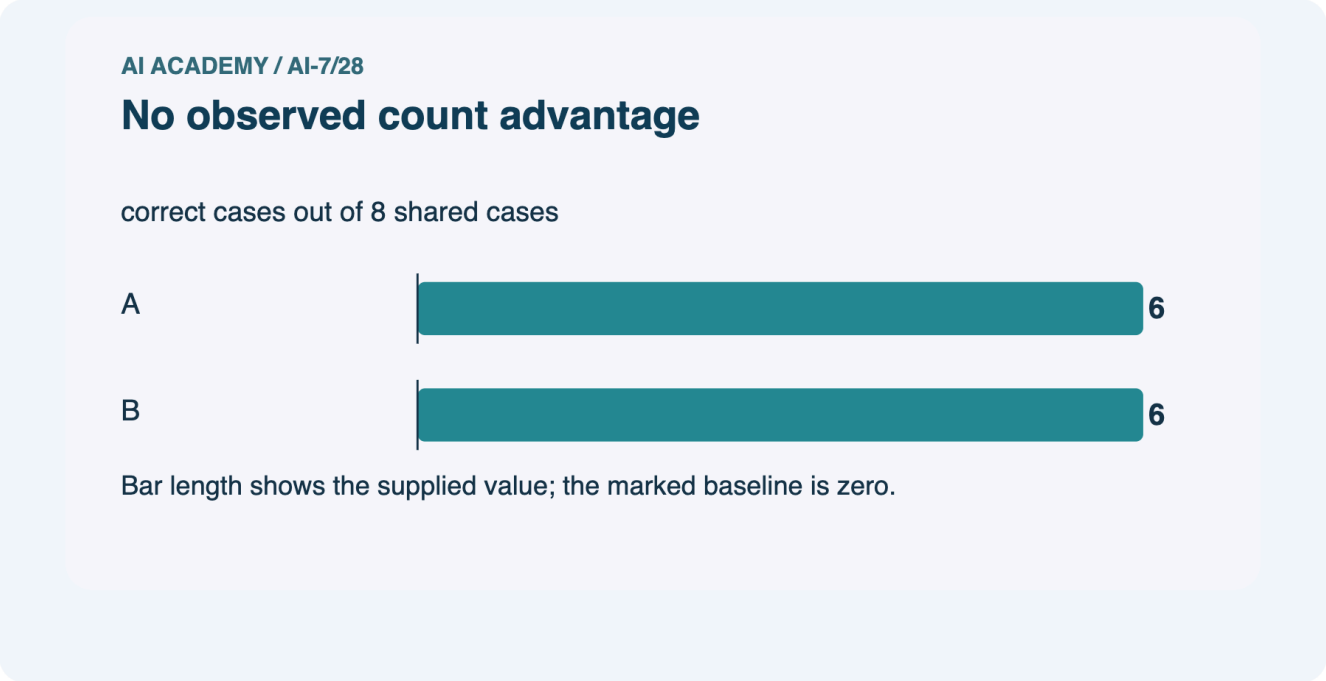
Follow the labelled picture with the matching explanation. Open the vector diagram

Recall the **prediction**: The claim was a higher correct count for B. → State the finding: Both versions score 75%; no observed count advantage appears here. → Revise cautiously: Report the unsupported **prediction** and a justified next test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. No observed count advantage



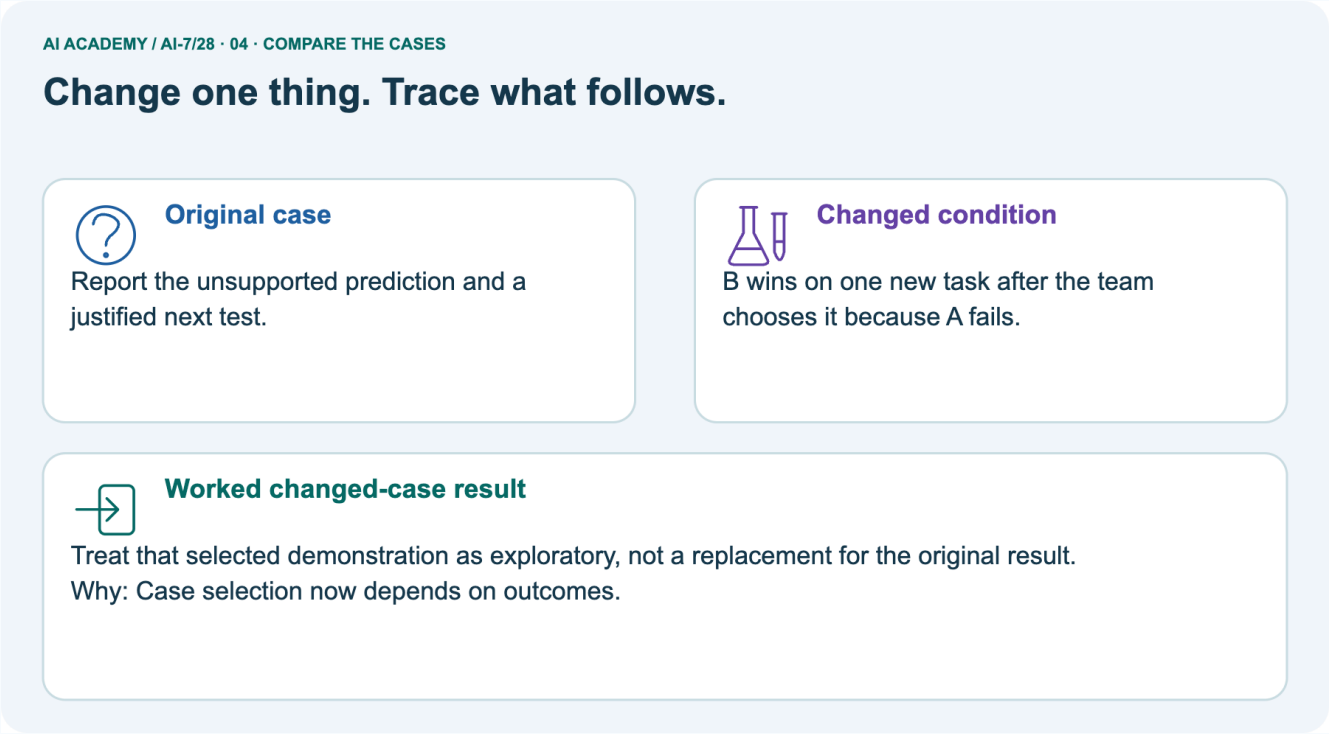
Follow the labelled picture with the matching explanation. Open the vector diagram

No observed count advantage. Units: correct cases out of 8 shared cases. A: 6. B: 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

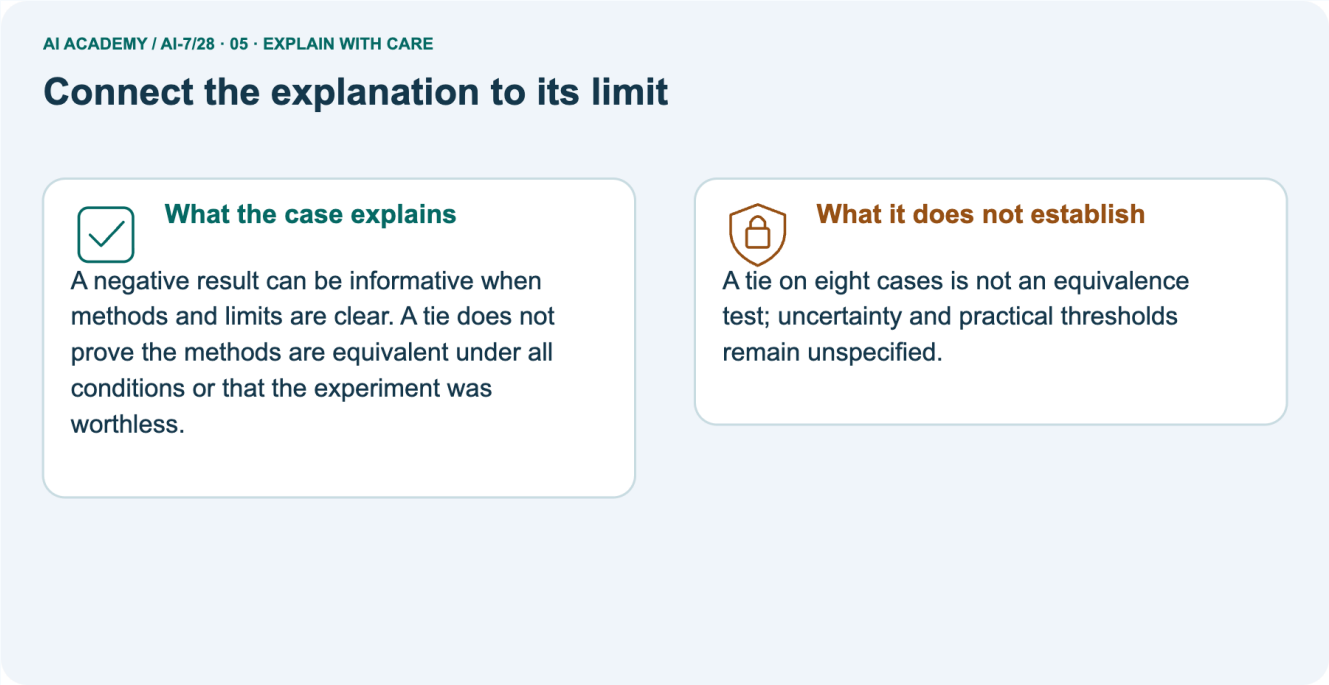
Original case: Report the unsupported **prediction** and a justified next test. Changed condition: B wins on one new task after the team chooses it because A fails.

Expected result: Treat that selected demonstration as exploratory, not a replacement for the original result. Reason: Case selection now depends on outcomes.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

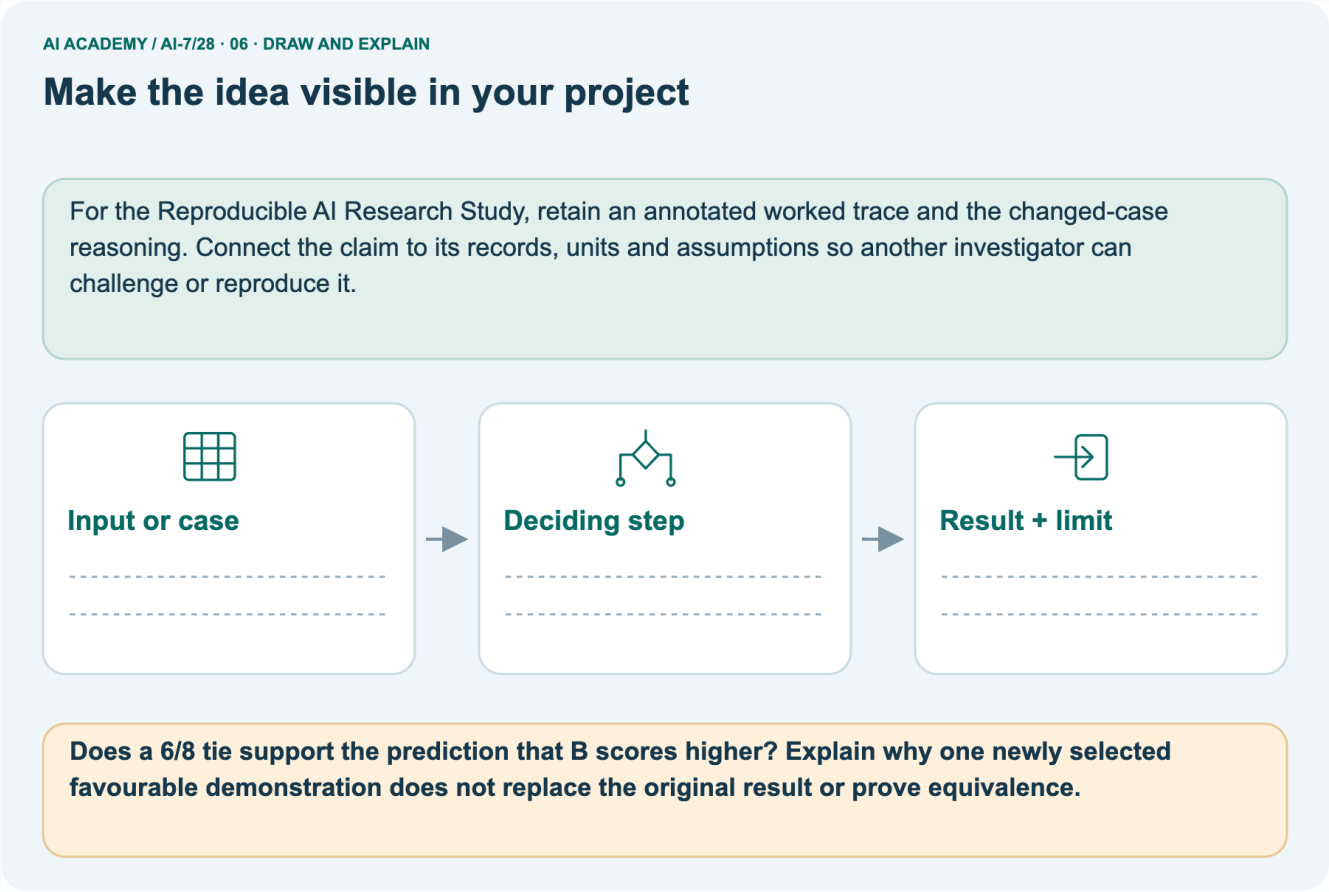
A **negative result** can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

Limit: A tie on eight cases is not an **equivalence** test; uncertainty and practical thresholds remain unspecified.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove **equivalence**.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.